

WORKFLOW SPECIFICATION REPORT

YPDC

Website & Portal System

Seven-Portal Digital Platform | Guest · Student · Member · Directorate · Faculty · Admin · Alumni

End-to-end business workflows, module inventory, approval chains and role-based access model

Prepared for: YPDC Executive Committee & Project Sponsors
Prepared by: YPDC Digital Secretariat
Version 1.0 (Draft for Approval) · 11 September 2026

GUEST

STUDENT

MEMBER

DIRECT.

FACULTY

ADMIN

ALUMNI

YPDC

Document Control

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Version	Date	Author	Summary of change
0.1	—	Digital Secretariat	Portal list captured from stakeholder input (7 portals).
0.9	—	Digital Secretariat	Module inventory, workflows and access matrix drafted.
1.0	11 Sep 2026	Digital Secretariat	Complete workflow specification issued for approval.

Approvals

Role	Name	Purpose of approval	Signature / date
Project Sponsor		Scope and budget acceptance	
Chairperson, YPDC		Organisational acceptance	
Faculty Focal Person		Academic and policy acceptance	
Technical Lead		Feasibility and delivery plan acceptance	

Note on the name “YPDC”

This document uses the abbreviation **YPDC** throughout. The full legal name is to be confirmed by the Executive Committee and substituted globally before external circulation.

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Executive Summary

YPDC currently runs its activities — membership, directorates, events, attendance, reporting and recognition — through a mixture of registers, spreadsheets, messaging groups and paper approvals. Information about a single member is therefore scattered, attendance is difficult to verify, approvals move at the speed of whoever is available, and the executive committee has no reliable data on which to plan.

This specification defines a single web platform organised into **seven portals**, one for each audience: **Guest/Public, Student, Member, Directorate, Faculty, Admin** and **Alumni**. The seven portals share one database, one identity system and one approval engine; they differ only in what a user may see and do. Together they cover **81 modules** and **11 documented end-to-end workflows**.

What this report delivers

- **A complete module inventory** — all 81 modules grouped by portal, each with its purpose, key functions and the data it owns.
- **End-to-end workflows** — step-by-step flows for onboarding, event lifecycle, approvals, attendance, volunteer management, certificates, performance, complaints and mentorship.
- **A role-based access model** — who can create, read, update, approve and delete in every module group.
- **A data model outline** — the 25 core entities that all portals read from and write to.
- **A phased delivery roadmap** — six phases from foundation to optimisation, with acceptance criteria.
- **Risks, assumptions and open decisions** that require an Executive Committee decision before development starts.

The seven portals at a glance

#	Portal	Primary users	Modules	Core value delivered
07	Guest / Public	Public, applicants, media, recruiters	14	Credibility, discovery and a single entry point for applications
01	Student	Enrolled students, prospective members	10	Low-friction participation, attendance and early recognition
02	Member	Inducted members, team leads	10	Task execution, duty record, performance visibility
03	Directorate	Directors, assistant directors	13	Planning, delegation, execution and evidence of work
04	Faculty	Focal persons, advisors	9	Academic oversight, endorsement and approval
05	Admin	Admin officers, content and data officers	15	Control room: identity, approvals, content, analytics
06	Alumni	Graduated members, mentors	10	Network, mentorship, careers and continued contribution
Total			81	

Decisions required from the Executive Committee

1. Confirm the full legal name of **YPDC** for the cover page and all templates.
2. Approve the role and permission matrix in Section 5, particularly who holds final approval authority.
3. Confirm the attendance verification method to be used on the ground (QR, geo-fence, offline code or manual).
4. Approve the performance weighting (tasks 40% / attendance 25% / events 20% / conduct 15%).
5. Confirm budget, hosting preference and the target launch date for Phase 2.

1. Introduction & Project Overview

1.1 Background

YPDC operates through directorates, each led by a director and supported by faculty focal persons, with students progressing into full membership and, after graduation, into an alumni body. Every layer generates records — who joined, who attended, what was delivered, who performed well — but none of these records currently live in one place.

The consequence is operational rather than cosmetic. Membership lists are reconciled by hand before every certificate run. Attendance disputes cannot be settled because the marking method leaves no evidence. An event proposal can sit with a single approver for weeks with no visibility for anyone else. And when the executive committee asks how many activities a directorate delivered last quarter, the answer has to be assembled manually.

A single platform with role-scoped portals removes all of these frictions at once, because the same record that a member updates is the record a director reviews, a faculty member endorses and an administrator reports on.

1.2 Objectives

#	Objective	How the platform achieves it	Measure of success
01	One trusted record per person	A single identity across all seven portals, with a verified profile and a permanent Member ID.	Zero duplicate member records at annual audit
02	Faster, transparent approvals	A configurable multi-level approval engine with SLA timers and auto-escalation.	Average approval cycle time reduced to under 5 working days
03	Verifiable attendance	Session-based marking with method, timestamp, geo and marker identity; reconciliation before lock.	Attendance disputes resolved from system evidence in under 48 hours
04	Evidence-based performance	Objective, task, attendance and event data aggregated into a published, appealable score.	100% of members able to see their score and its components
05	Professional public presence	A CMS-driven public portal for content, events, gallery, publications and applications.	All content published by the admin team without developer involvement
06	Recognition at scale	Rule-based certificate issuance with unique serials and public QR verification.	Certificates issued within one working day of activity closure
07	Data for decisions	Analytics on membership funnel, event outcomes, attendance and complaints.	Quarterly executive report generated from the system without manual collation
08	Lifelong relationship	An alumni portal for network, mentorship, careers and continued contribution.	Active mentorship pairs and alumni-posted opportunities each quarter

1.3 Scope

In scope

- Seven role-based portals sharing one database, one authentication service and one approval engine.
- The 81 functional modules inventoried in Section 8, including their workflows and data.
- Cross-portal workflows: onboarding, event lifecycle, approvals, attendance, volunteering, certificates, performance, complaints and mentorship.
- The role and permission model, audit logging, notifications and public certificate verification.
- Content management for the public website, including news, gallery, publications and FAQs.

Out of scope (for this phase)

- Native mobile applications — the platform is responsive web first; native apps may follow once the workflow is stable.
- Full accounting and payroll systems — only budget capture and utilisation against events are in scope.

- Academic result management of the host institution — academic data is stored for context only.
- Biometric attendance hardware integration — the data model allows it, but no hardware is procured in this phase.
- Payment gateway integration — membership fees are recorded as paid/unpaid; online collection is a later phase.

1.4 Key stakeholders

Stakeholder	Interest	Primary portal	What they need from this project
Executive Committee / Patron	Governance, reputation, assurance	Admin (reports)	Reliable reports, control over approvals and risk visibility
Directors & Assistant Directors	Delivery of directorate objectives	Directorate	Planning, delegation and evidence with minimal paperwork
Faculty focal persons	Academic legitimacy and student welfare	Faculty	A light approval queue with full context, and oversight flags
Members & students	Participation, recognition, growth	Member / Student	Clarity on duties, fair attendance and timely certificates
Admin team	Day-to-day operation of the platform	Admin	Control over users, content and approvals without developer help
Alumni	Connection, mentorship, giving back	Alumni	A current network and simple ways to contribute
Public & media	Transparency and information	Guest	Accurate public information and easy application routes

2. Platform Architecture & Portal Model

2.1 One platform, seven views

The seven portals are **not seven systems**. They are seven permission-filtered views over a single data store. A record written by a member in the Member Portal is instantly visible — subject to permission — in the Directorate, Faculty, Admin and Alumni views. This single-source-of-truth design is what makes approvals fast, attendance verifiable and reporting automatic.

Layer	Responsibility	Key components	Notes
1. Presentation	What each audience sees and can do	Seven portal shells (public site + six authenticated workspaces), responsive layouts, role-driven navigation and dashboards	One codebase; the portal rendered is decided by the user's roles at login
2. Application / API	Business rules and workflow orchestration	AuthN/AuthZ service, approval engine, certificate rules engine, attendance service, notification service, CMS, analytics	All portals call the same API; no portal owns private logic
3. Data	The single source of truth	Relational store for the 25 core entities, object storage for files and media, audit log store, cache	Soft delete and full history on every regulated entity
4. Infrastructure	Availability, security and delivery	HTTPS everywhere, automated backup, CI/CD pipeline, monitoring and alerting, staging and production environments	Hosting decision required — see Section 16, open items

2.2 Shared platform services

Eight services are used by every portal. Building them once, centrally, is what prevents the seven portals from drifting into seven inconsistent experiences.

Service	What it does	Used by
Identity & authentication	Single account per person; login by e-mail or Member ID with OTP/password; optional 2FA for staff roles; session and device management.	All authenticated portals
Authorisation (roles & permissions)	Module × action permission sets attached to roles; evaluated on every request and used to render navigation.	All portals
Approval engine	Configurable multi-level chains, SLA timers, auto-escalation, delegation and immutable decision history.	Student, Member, Directorate, Faculty, Admin, Alumni
Notification service	In-app, e-mail and SMS delivery from templates; per-user preferences; digest and scheduling; delivery analytics.	All portals
Attendance service	Session creation, multi-channel marking, anomaly detection, reconciliation, condonation and period lock.	Student, Member, Directorate, Admin
Certificate engine	Rule evaluation, template rendering, unique serial and QR generation, issuance, revocation and public verification.	Student, Member, Directorate, Admin, Alumni
Content management (CMS)	Pages, news, FAQs, projects, gallery albums and publications with draft → review → publish and versioning.	Admin writes; Guest reads
Audit & analytics	Immutable audit log of regulated actions; KPI computation; dashboards and scheduled exports.	Admin primarily; scoped views elsewhere

2.3 How the portals relate

The diagram below shows the flow of people and records between portals. Everything on the left enters through the public surface; everything on the right is governed centrally.

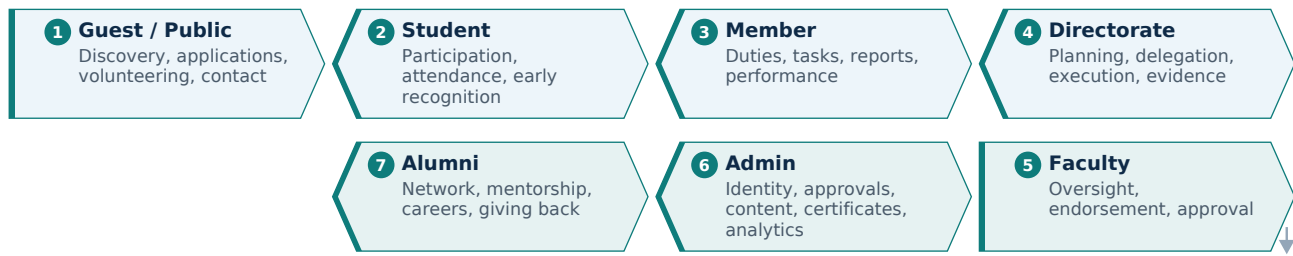


Figure 1 — Portal progression: a visitor becomes a student, a member, a contributor, and eventually an alumnus, while the Directorate, Faculty and Admin portals govern the journey throughout.

2.4 Module inventory summary

Portal	Modules	Modules that write data	Modules that primarily approve / review	Depends on approval engine
Guest / Public	14	4 (forms, contact, FAQ vote)	0	Yes — applications
Student	10	6	0	Yes — membership, complaints
Member	10	7	0	Yes — condonation, certificates
Directorate	13	11	2 (attendance, activity reports)	Yes — event proposals
Faculty	9	1 (feedback)	6	Yes — approvals module
Admin	15	13	10	Yes — final authority
Alumni	10	8	0	Yes — mentorship, postings
Total	81	50	18	

3. User Roles & Access Model

3.1 Role definitions

A **role** is a bundle of permissions (module × action). A user may hold several roles at once — for example a member who is also a team lead, or a faculty member who is also an alumni. The portal shown after login is determined by the user's highest privilege role, with a portal switcher available where multiple apply.

Role	Who holds it	Portal	Scope of data visible	Cannot do
Guest	Any public visitor	Guest / Public	Published content only	See any member personal data
Student	Enrolled student / prospective member	Student	Own records; public events	Approve anything; see other students' records
Member	Inducted member with Member ID	Member	Own records; own directorate's public plan and roster	Edit objectives or work plan
Team Lead	Member with a sub-team	Member + Directorate (limited)	Own sub-team's tasks and attendance	Approve event proposals or certificates
Assistant Director	Deputy head of a directorate	Directorate	Own directorate	Final approval on budget and policy
Director	Head of a directorate	Directorate	Own directorate, full	Manage users outside the directorate; change permissions
Faculty	Focal person / advisor	Faculty	Assigned directorates and student groups	Administer users, roles or website content
Alumni	Graduated member	Alumni	Own profile; opted-in network	See current members' private records
Admin Officer	Operations staff	Admin	All records	Change own permission set (segregation of duty)
Content Manager	Communications team	Admin (content modules)	Content, media, publications	Approve membership or edit users
Super Admin	Technical / secretariat head	Admin (full)	Everything including roles and audit log	— (all actions are still audit-logged)

3.2 Permission matrix

The matrix below is the authoritative summary of access rights. It is implemented as permission sets in the Admin Portal (Roles & Permissions module) and enforced on every API request — not merely hidden in the user interface.

Legend: ● full rights (create / update / delete) ○ own or scoped records only ▲ review / approve only – no access

Module group	Guest	Student	Member	A/D	Director	Faculty	Admin	Alumni
Public website content	–	R	R	R	R	R	●	R
Applications (membership / volunteer)	● own	● own	–	▲	▲	▲	●	–
Own profile	–	●	●	●	●	●	●	●
Other users' profiles	–	–	○ roster	○ unit	○ unit	○ assigned	●	○ opted-in
Membership lifecycle	–	○	○	▲	▲	▲	●	–
Directorate profile & structure	R	R	R	○	●	R	●	R
Objectives & work plan	–	–	R	○	●	▲	●	–
Tasks	–	–	○	○	●	R	●	–
Events & activities	R	R	R	○	●	▲	●	R
Event proposals	–	–	○	▲	●	▲	●	–

Module group	Guest	Student	Member	A/D	Director	Faculty	Admin	Alumni
Registrations & duty roster	● own	● own	● own	● unit	● unit	R	●	● own
Attendance	-	○	○	● unit	● unit	R	● audit	-
Activity reports	-	-	● own	▲	▲	▲	●	-
Certificates	▲ verify	○	○	▲ recommend	▲ recommend	▲	● issue	○
Performance	-	R rules	○	● unit	● unit	▲	●	-
Approvals queue	-	-	-	▲	▲	▲	●	-
Feedback & complaints	● raise	● raise	● raise	○ resolve	○ resolve	○ review	● triage	● raise
User, role & permission admin	-	-	-	-	-	-	●	-
Reports & analytics	-	-	○ own	○ unit	○ unit	○ assigned	●	-
Alumni network & mentorship	-	-	○ mentee	-	-	R	●	● mentor

Two rules that must never be relaxed

Segregation of duty: no user may approve a request they initiated, and no user may modify their own permission set. Both are enforced server-side.

Least privilege by default: every new account starts with read access to its own records only. Additional rights are granted explicitly and reviewed each term.

4. Core End-to-End Workflows

This section documents the eleven workflows that cross portal boundaries. Each is written as it will be implemented: the trigger, every step, every decision point, both outcomes of each decision, and the record that is created. Sections 8.1 to 8.7 then describe the internal journey within each individual portal.

#	Workflow	Triggered by	Portals involved	Primary owner
W1	Membership / volunteer onboarding	Public application form	Guest, Admin, Faculty, Directorate, Member	Admin
W2	Event lifecycle	Directorate event proposal	Directorate, Faculty, Admin, Student, Member, Guest, Alumni	Director
W3	Generic approval engine	Any submitted form	All authenticated portals	Admin
W4	Attendance capture to certificate	Session creation	Directorate, Admin, Student, Member	Director
W5	Volunteer registration to recognition	Public volunteer form	Guest, Admin, Directorate, Member	Admin
W6	Feedback & complaint handling	Feedback / contact submission	All portals, Admin	Admin
W7	Alumni mentorship cycle	Mentorship request	Member, Alumni, Admin	Alumni lead
W8	Certificate issuance & verification	Achievement trigger	Directorate, Admin, Member, Student, Guest	Admin
W9	Performance management cycle	Period start (monthly / quarterly)	Directorate, Member, Faculty, Admin	Director
W10	Content publication to public site	Content created by admin	Admin, Guest	Content Manager
W11	Notification & communication flow	Any system or human event	All portals	Admin

4.1 W1 — Membership & volunteer onboarding

The single most important workflow in the platform: it converts a stranger on the public website into a tracked, portal-enabled member with a permanent ID. It is deliberately staged so that an incomplete application never reaches a human reviewer.



Figure 2 — Membership and volunteer onboarding. Two decision gates keep reviewer time focused on genuine candidates; every outcome is notified to the applicant with a reference ID.

Service levels and rules

Control point	Rule	Owner
Acknowledgement	Reference ID issued instantly on submission; e-mail + SMS within 1 minute.	System
Completeness check	Automated; incomplete files returned with the exact missing items.	System
Auto-close	Incomplete applications close after 14 days without resubmission.	System
Screening SLA	Scheduled within 10 working days of a complete application.	Directorate
Final approval SLA	Within 3 working days of a positive recommendation.	Admin
Cooling period	A declined applicant may reapply after the defined period, unless declined on conduct grounds.	Admin
Duplicate control	Checked against national ID / institutional roll number / phone before approval.	System + Admin

4.2 W2 — Event lifecycle

Events are the main delivery mechanism of YPDC, so the event lifecycle touches every portal. The workflow separates approval (four levels, each with a distinct purpose) from execution (three parallel tracks) and closure (reporting, certificates and analytics).

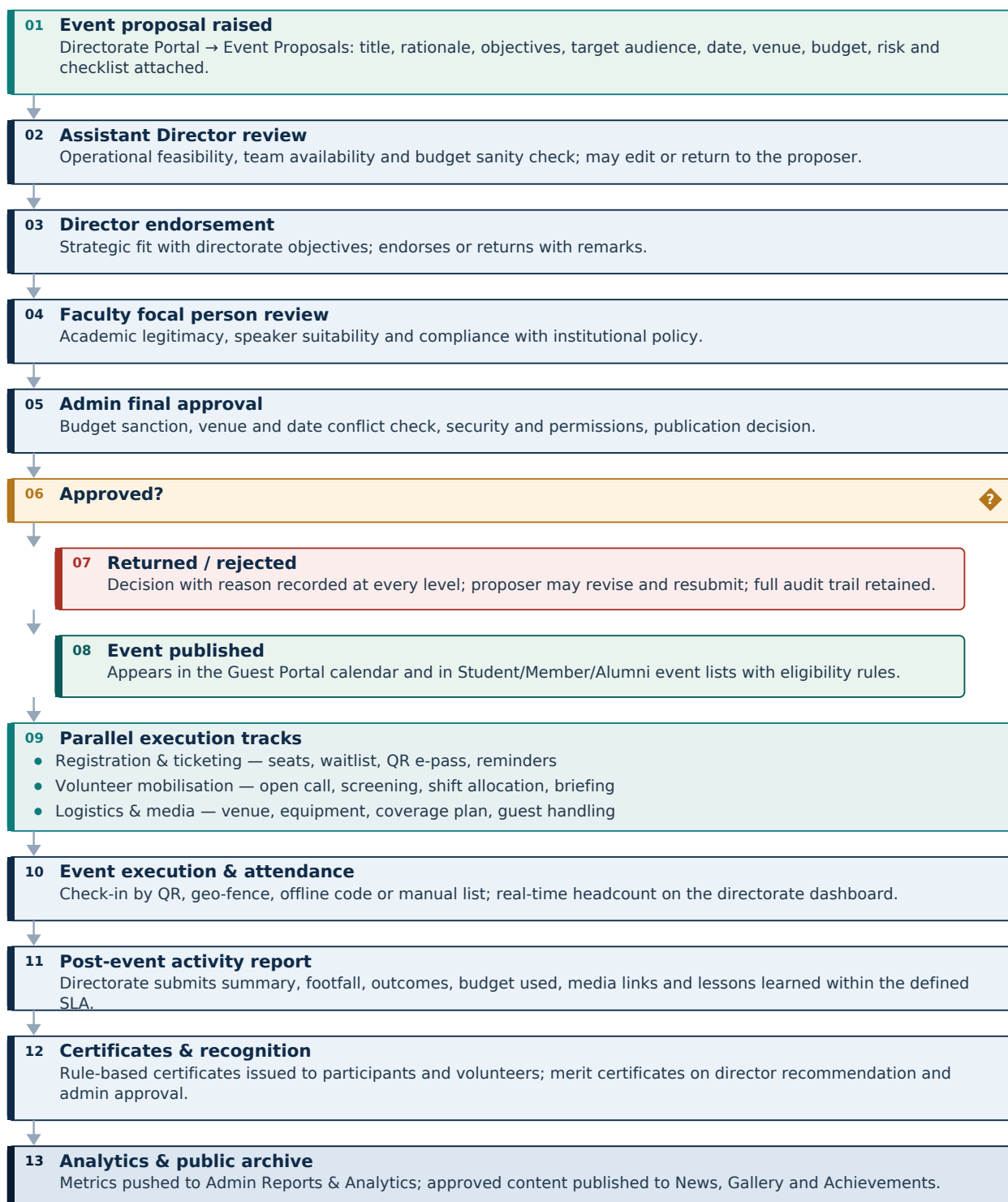


Figure 3 — Event lifecycle from proposal to public archive. Approval levels are sequential; execution tracks run in parallel; closure is not optional.

Approval levels — what each one actually decides

Level	Approver	Decision focus	Typical SLA	Can they edit the proposal?
1	Assistant Director	Operational feasibility, team and date availability	2 working days	Yes, minor edits
2	Director	Strategic fit with directorate objectives and annual plan	2 working days	Yes, with remark
3	Faculty focal person	Academic legitimacy, speaker suitability, institutional policy	3 working days	Comment only
4	Admin	Budget sanction, venue and date conflict, security, publication	3 working days	No — approve or return

Closure requirements (mandatory before an event can be marked complete)

- Attendance reconciled and confirmed for every session of the event.
- Activity report submitted: summary, actual footfall, outcomes, budget used, media links, lessons learned.
- Certificates drafted and sent for approval where the event type carries a certificate rule.
- Gallery and news content submitted for publication, with consent flags set per image.
- Budget variance recorded, and any unused allocation released back to the directorate.

4.3 W3 — The generic approval engine

Rather than writing a separate approval routine for every form type, the platform implements one configurable engine. A new form type (for example a budget request or a condonation appeal) is added by configuration — defining its levels, approvers, thresholds and SLAs — without new development.

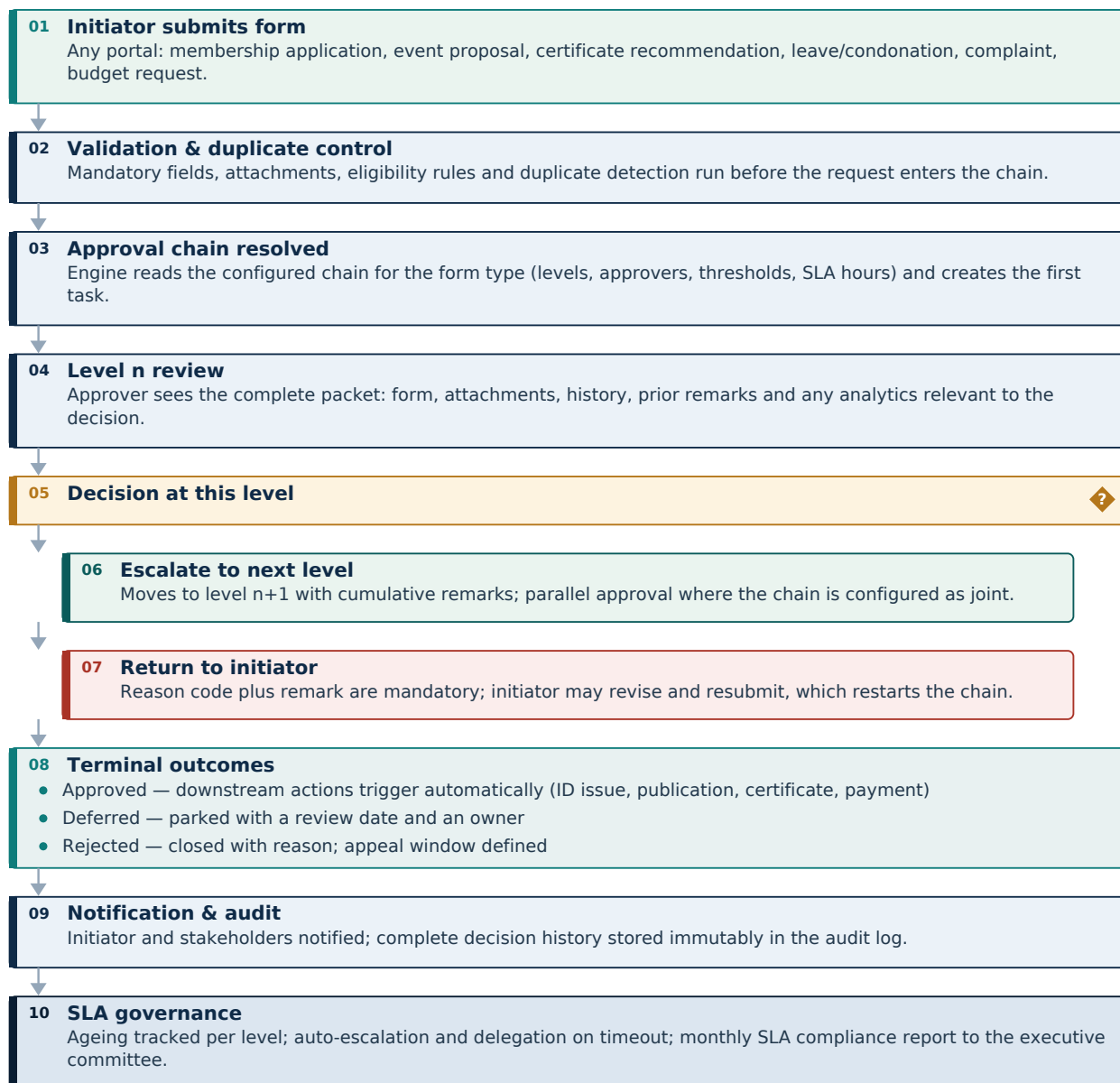


Figure 4 — Generic approval engine. The same engine serves membership, event proposals, certificates, condonation, complaints and budget requests.

Configured approval chains

Form type	Approval chain	Parallel / sequential	SLA (total)	Escalation on timeout
Membership application	Director → Faculty → Admin	Sequential	10 working days	To Super Admin
Event proposal	Assistant Director → Director → Faculty → Admin	Sequential	10 working days	To Super Admin
Merit certificate	Director → Admin	Sequential	5 working days	To Super Admin
Attendance condonation	Director → Admin	Sequential	5 working days	To Faculty focal person

Form type	Approval chain	Parallel / sequential	SLA (total)	Escalation on timeout
Budget request	Director → Admin (threshold-based second approver above limit)	Sequential	7 working days	To Project Sponsor
Complaint (P2-P4)	Admin triage → Assigned owner → Admin closure	Sequential	By priority	To grievance committee
Complaint (P1)	Admin → Grievance committee → Patron	Sequential	48 hours	Immediate to Patron
Alumni job / support posting	Admin moderation only	Single level	2 working days	To Content Manager
Mentorship match	Mentor acceptance → Alumni lead confirmation	Sequential	7 working days	Re-match to next candidate

4.4 W4 — Attendance: capture to certificate

Attendance is the most disputed data in any membership organisation. This workflow makes every mark traceable to a method, a time, a location and a person, and inserts a reconciliation step before any period is locked for reporting.

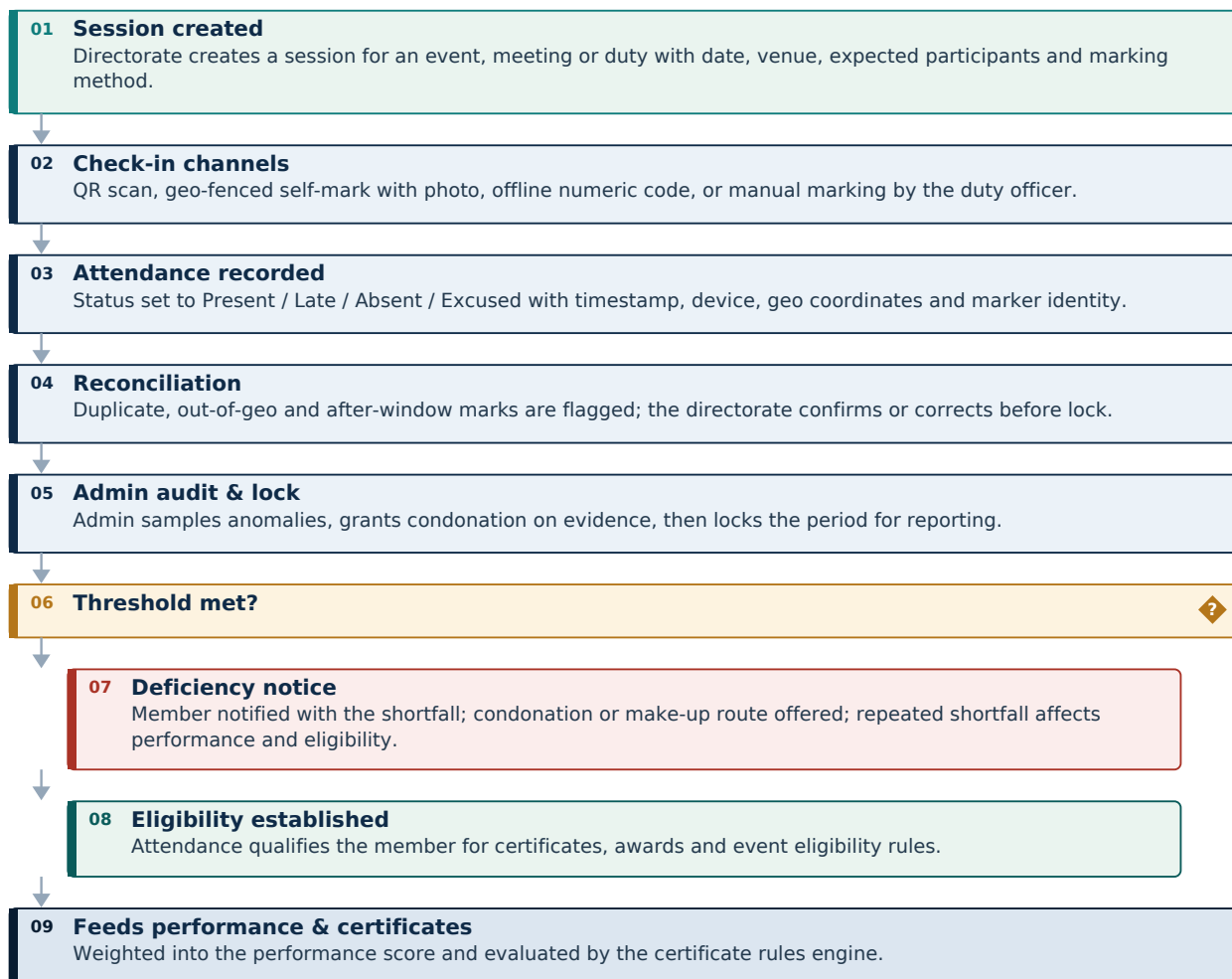


Figure 5 — Attendance workflow. Note the reconciliation gate before the period lock: once locked, corrections require an admin-level adjustment with a recorded reason.

Marking methods and their controls

Method	How it works	Fraud controls	Best used for
QR scan	A session QR is displayed or printed; the participant scans it from the portal.	Time-boxed QR refresh; single scan per user; scan location logged.	Large events, seminars
Geo-fenced self-mark	Participant marks attendance from a phone within a radius of the venue.	Radius check, optional photo, device and timestamp recorded.	Outdoor activities, field visits
Offline numeric code	Duty officer announces a rotating code; participants enter it.	Short code life; one use per user; attempts logged.	Poor-connectivity venues
Manual list	Duty officer marks a printed or on-screen list.	Marker identity recorded; reconciliation and admin sampling; two-person confirmation for large sessions.	Small meetings, duty rosters

Thresholds are configurable, not hard-coded

Minimum attendance percentages, deficiency warning levels, condonation limits and the attendance lock window are stored as configuration in the Admin Portal so that policy changes never require a code release. The default starting point proposed here is: warning at 80%, deficiency below 75%, condonation limited to two per year on evidence.

4.5 W5 — Volunteer registration to recognition



Figure 6 — Volunteer journey. The volunteer never leaves the funnel: even a declined assignment keeps the profile warm in the talent pool.

4.6 W6 — Feedback & complaint handling

Complaints are the platform's early-warning system. The workflow guarantees that no complaint disappears: every ticket has an owner, an SLA, a recorded resolution and a satisfaction rating, and sensitive matters escalate immediately regardless of queue load.

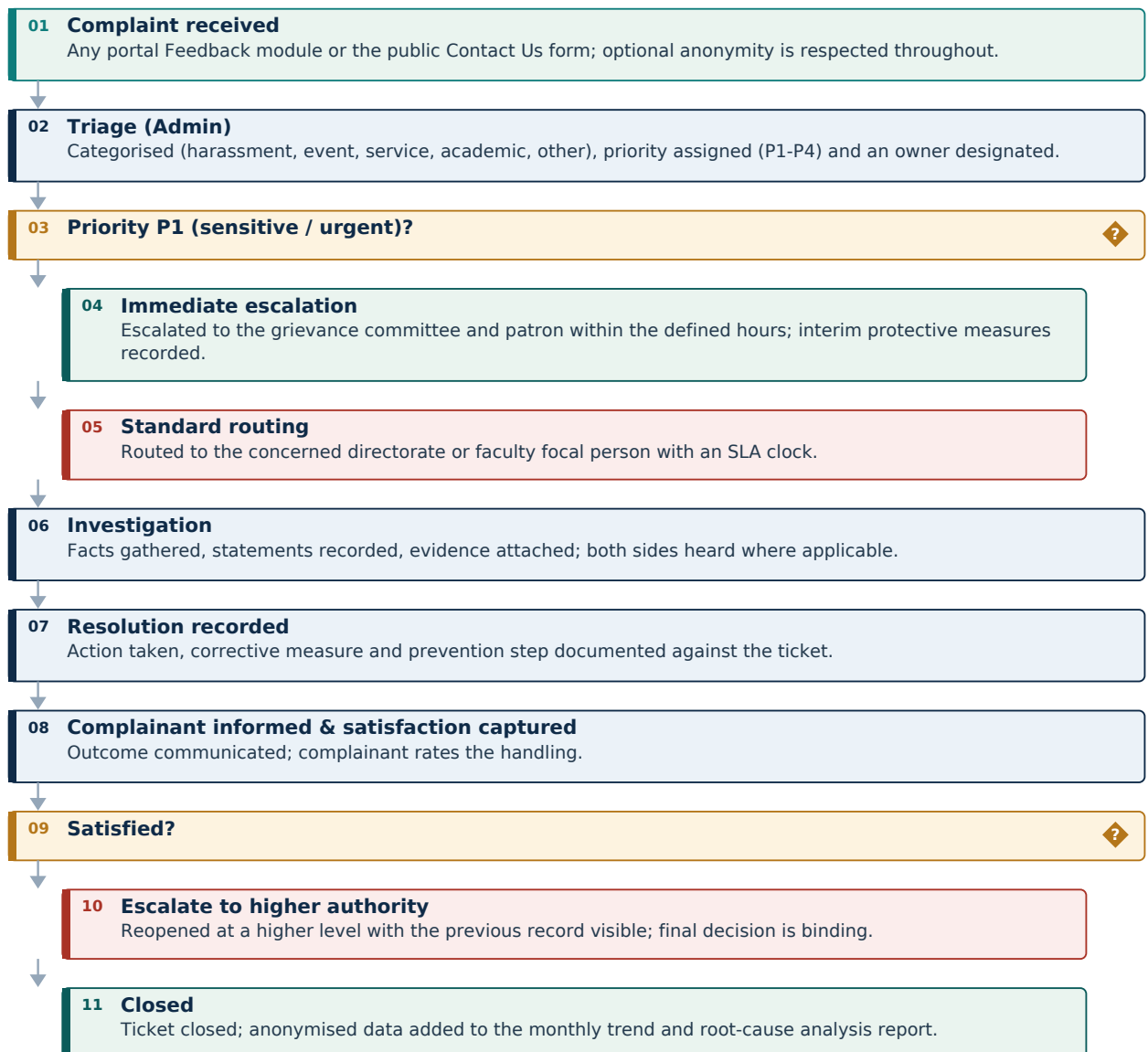


Figure 7 — Complaint handling with a dedicated fast path for P1 (sensitive or urgent) matters and a re-opening route when the complainant is not satisfied.

Priorit y	Definition	Response	Resolution target	Escalation
P1	Safety, harassment, discrimination, legal exposure	Immediate	48 hours	Grievance committee + Patron
P2	Significant service failure affecting a group	1 working day	5 working days	Admin head
P3	Individual service or event issue	2 working days	10 working days	Directorate head
P4	Suggestion, minor observation, FAQ gap	3 working days	Next monthly review	—

4.7 W7 — Alumni mentorship cycle



Figure 8 — Mentorship cycle. Matching is proposal-based and requires mutual acceptance; sessions are logged so that contribution can be recognised.

4.8 W8 — Certificate issuance & public verification

A certificate is only valuable if a third party can trust it. This workflow separates the automatic rule evaluation from the human approval, assigns a unique serial at the point of issue, and exposes a public verification endpoint so that employers and institutions can confirm authenticity in seconds.



Figure 9 — Certificate issuance and verification. Revocation is immediate and always reflected on the public verification endpoint.

Certificate type	Trigger	Approval needed	Validity	Serial pattern
Participation	Confirmed registration + verified attendance	None (rule-based auto-issue)	Permanent	YPDC-PRT-YYYY-NNNNN
Volunteer service	Completed duty with a submitted duty report	None (rule-based auto-issue)	Permanent	YPDC-VOL-YYYY-NNNNN
Completion (training / programme)	Attendance threshold + assessment where applicable	Director recommendation	Permanent	YPDC-CMP-YYYY-NNNNN
Merit / excellence	Performance score or competition result	Director + Admin	Permanent	YPDC-MRT-YYYY-NNNNN
Appointment / office bearer	Appointment to a role	Admin	Duration of tenure	YPDC-APPT-YYYY-NNNNN
Alumni contribution	Mentorship hours or support delivered	Alumni lead + Admin	Permanent	YPDC-ALM-YYYY-NNNNN

4.9 W9 — Performance management cycle

Performance in YPDC must be objective and visible, otherwise it becomes a source of grievance rather than a tool for growth. Every component of the score is derived from records the member themselves created, and every score carries an appeal route.

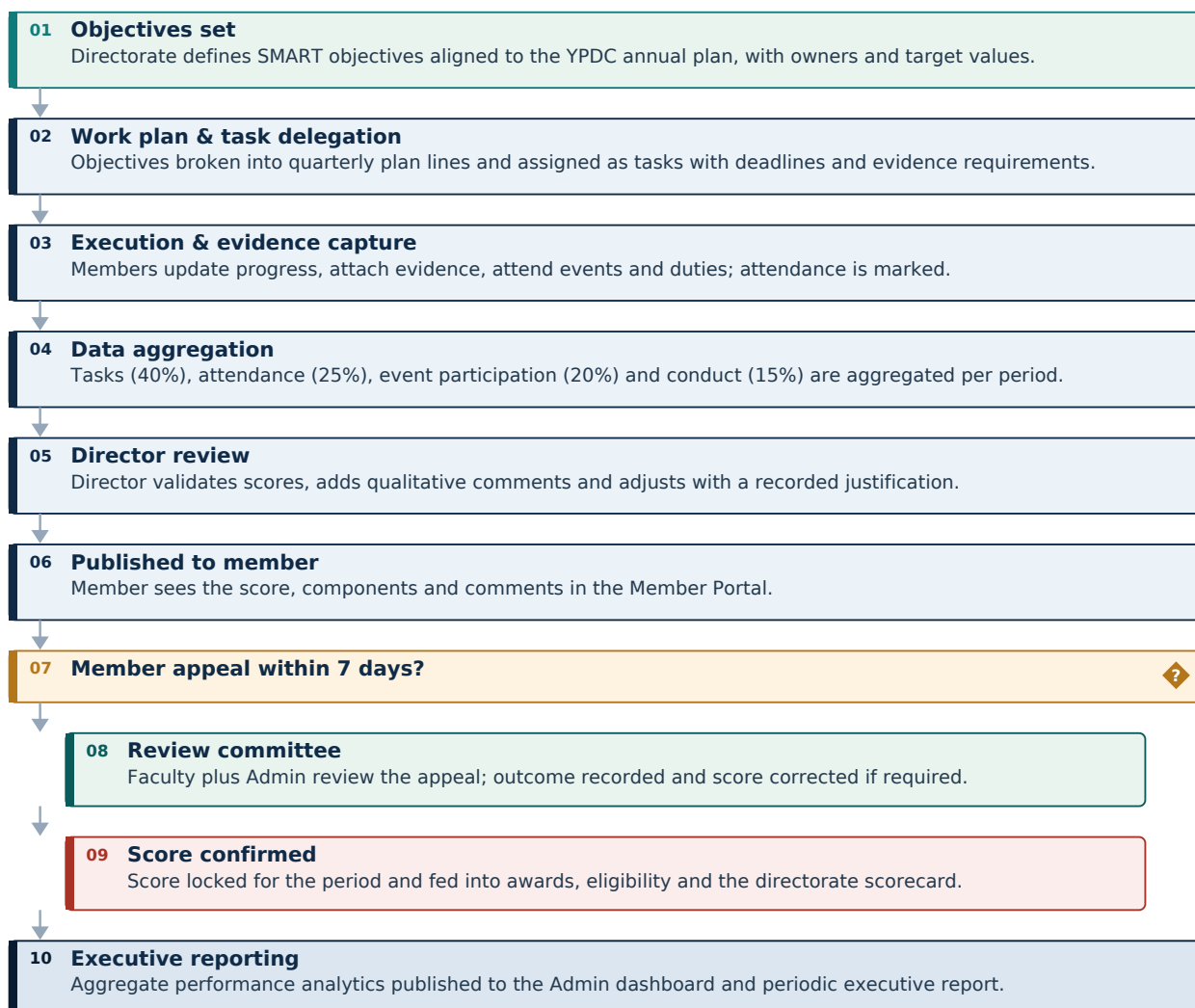


Figure 10 — Performance cycle. The appeal window is the key control: it converts a dispute into a documented, reviewable decision.

Component	Weight	Data source	Notes
Tasks completed	40%	Task module — status, timeliness and reviewer rating	Late submission scored partially; overdue and unsubmitted scored zero
Attendance	25%	Attendance service — reconciled and locked records	Category-wise weighting (duties weigh more than general sessions)
Event & activity participation	20%	Event registrations, duty roster, activity reports	Leadership roles (convener, team lead) earn a multiplier
Conduct & contribution	15%	Director comments, peer feedback, complaints upheld	Any upheld P1/P2 complaint reduces this component to zero for the period

Fairness controls built into the score

Members see the score **and its components**, never a single opaque number.

A director's manual adjustment **requires a written justification** that the member can read.

The **7-day appeal window** is enforced by the system: after it closes, the period is locked.

Periods in which a member had no assigned tasks are **excluded** rather than scored zero.

4.10 W10 — Content publication to the public site

Everything the public sees is managed through the Admin Portal's Website Content module. The workflow enforces review before publication, keeps a version history, and never exposes member data without an explicit consent flag.



Figure 11 — Content publication workflow for the Guest/Public Portal.

4.11 W11 — Notification & communication flow

Notifications are the mechanism that keeps every other workflow moving. They are always template-driven and preference-controlled, so that a member is never spammed and a deadline is never missed silently.

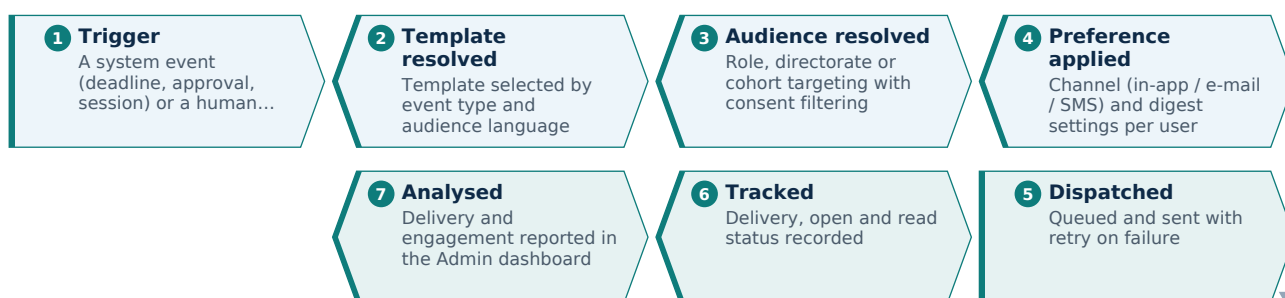


Figure 12 — Notification flow from trigger to analytics.

Notification category	Examples	Default channels	Who controls it
Account & security	Welcome, password reset, new device login, role change	E-mail (mandatory) + in-app	System
Membership	Application received, screening date, approved, declined, expiring	E-mail + SMS + in-app	Admin
Tasks & duties	Task assigned, deadline T-3 / T-1 / overdue, duty roster published	In-app + e-mail	Directorate
Approvals	New item in queue, SLA warning, decision recorded	In-app + e-mail	System
Events	Published, registration confirmed, reminder, cancelled, report due	In-app + e-mail + SMS	Directorate / Admin
Attendance & certificates	Deficiency warning, condonation decision, certificate issued	In-app + e-mail	System / Admin
Complaints	Ticket received, assigned, resolved, satisfaction request	E-mail + in-app	Admin
Broadcasts	Announcements, campaigns, emergency notices	Selected by sender	Admin
Alumni	Mentorship match, opportunity posted, reunion invite	E-mail + in-app	Alumni lead

5. Portal-by-Portal Module Specification

This section is the functional core of the report. For each of the seven portals it states the purpose, the audience, the access level, the main user journey as a flow, and a complete module inventory with purpose, key functions and the data each module owns. The order follows the user's progression through the organisation, beginning with the public surface.

Portal	Modules covered in this section
07 • Guest / Public Portal	About YPDC, Vision & Mission, Leadership, Directorates, Projects, Events, News & Updates, Achievements, Gallery, Publications, Membership, Volunteer Registration, Contact Us, FAQs
01 • Student Portal	Profile, Membership, Events, Event Registration, Volunteer Opportunities, Attendance, Certificates, Achievements, Notifications, Feedback
02 • Member Portal	Profile & Member ID, Directorate, Responsibilities, Tasks, Events / Duties, Attendance, Certificates, Performance, Reports, Notifications
03 • Directorate Portal	Directorate Profile, Director / Assistant Director, Team Members, Objectives, Work Plan, Tasks, Events & Activities, Event Proposals, Attendance, Activity Reports, Achievements, Documents, Performance
04 • Faculty Portal	Faculty Profile, Department / Designation, YPDC Role, Events, Student / Member Activities, Approvals, Reports, Feedback, Notifications
05 • Admin Portal	Dashboard, User Management, Member Management, Directorate Management, Faculty Management, Events Management, Applications & Approvals, Attendance, Certificates, Reports & Analytics, Notifications, Website Content, Documents, Feedback / Complaints, Roles & Permissions
06 • Alumni Portal	Alumni Profile, Academic Information, Professional Information, Skills, Alumni Network, Mentorship, Events, Achievements, Career Opportunities, Volunteer / Support Opportunities

PORTAL 07

Guest / Public Portal

Unauthenticated public interface - the discovery, credibility and entry point of YPDC

Portal Specification — Guest / Public Portal

Purpose

The Guest/Public Portal is the only fully public surface of the platform. It converts anonymous visitors into applicants, volunteers, partners and audiences. Everything here is published by the Admin Portal (Website Content module) and cached for performance; no personal data of members is exposed without explicit consent.

Attribute	Detail
Target users	Public visitors, prospective students and members, parents, media, partner institutions, recruiters, alumni prospects.
Access level	Public - no login required. Write actions limited to forms (membership, volunteer, contact).
Modules	14 (inventoried below)

Main user journey

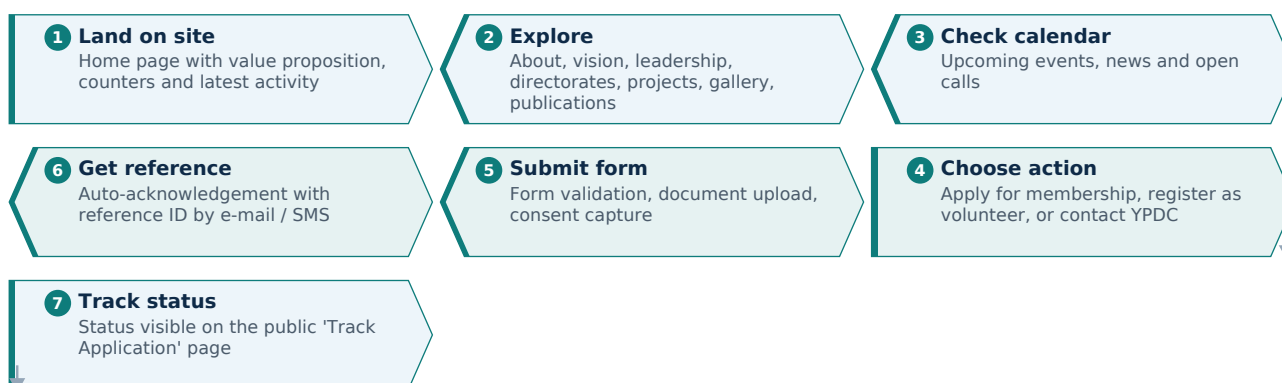


Figure 13 — Guest / Public Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
About YPDC	Introduce the organisation, its history, mandate and structure.	Static CMS page with rich media; history timeline; organogram; downloadable profile; last-updated stamp and page owner.	ContentPage, MediaAsset
Vision & Mission	Publish strategic intent and core values.	Vision / mission / values blocks; strategic pillars; annual theme; print-ready version.	ContentPage
Leadership	Show governance and patronage.	Patron, president and office-bearer profiles with photo, tenure and portfolio; succession chronology.	User, Profile, ContentPage
Directorates	Present all directorates as functional units.	Directorate cards (name, mandate, director, team size); link to objectives and public activities; contact focal person.	Directorate, Profile
Projects	Showcase ongoing and completed projects.	Project listing with status, timeline, progress %, partner and impact metrics; detail page with gallery and reports.	Project, ActivityReport

Module	Purpose	Key functions / rules	Primary data entities
Events	Public event calendar and event detail pages.	Upcoming / past split; filters (type, directorate, venue, month); event page with agenda, speakers, map; registration link where seats are public.	Event, Registration
News & Updates	Announcements, press releases and notices.	Chronological feed; categories and tags; pinned notices; RSS / share links; publication date and author.	ContentPage, MediaAsset
Achievements	Celebrate organisational and member achievements.	Achievement cards (title, member/directorate, date, awarding body); consent-gated publishing; yearly archive.	Achievement
Gallery	Visual record of activities.	Albums by event; lazy loading; download with watermark; consent flag per image; moderation queue before publish.	MediaAsset, Album
Publications	Repository of reports, newsletters and research.	Document listing with type, year and file size; preview and download counters; access level (public / member-only).	Document
Membership	Explain membership and accept applications.	Tiers, eligibility, benefits, fee schedule and process diagram; online application form with document upload; application reference number issued instantly.	Application, Document
Volunteer Registration	Capture public volunteers for YPDC activities.	Multi-step form (personal, skills, availability, interests); CV upload; consent for data processing; auto-acknowledgement and reference ID.	Application, User
Contact Us	Single entry point for enquiries.	Enquiry form with category routing; department emails; office address and map; working hours; ticket ID returned to sender.	Ticket
FAQs	Self-service answers to repeated questions.	Categorised accordion; search; helpful / not-helpful vote; unanswered questions promoted to admin for new FAQ creation.	ContentPage, Ticket

PORTAL 01

Student Portal

Authenticated space for enrolled students - membership, participation, attendance and recognition

Portal Specification — Student Portal

Purpose

The Student Portal is the first authenticated experience of a YPDC participant. It is designed to be low-friction: a student logs in with an institutional or e-mail account, completes a profile, tracks membership status, registers for events and volunteer drives, and collects attendance, certificates and achievements that later support a full membership application.

Attribute	Detail
Target users	Enrolled students of partner institutions; prospective members.
Access level	Authenticated. Reads own records only; writes limited to own profile, registrations and feedback.
Modules	10 (inventoried below)

Main user journey

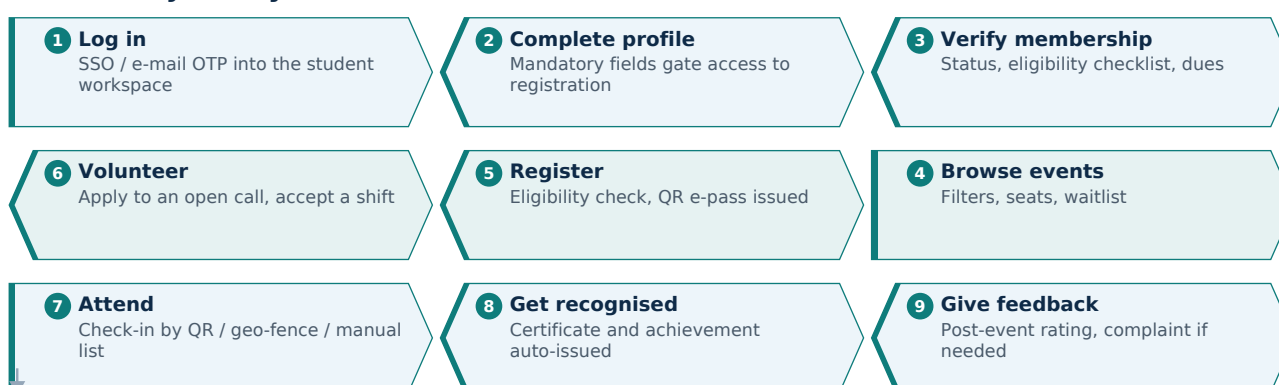


Figure 14 — Student Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
Profile	Single source of truth for student identity and contact.	Personal, academic and contact details; photo upload with auto-crop; institution, programme, semester; completion score; change history with audit trail.	User, Profile
Membership	Show and progress the student's membership status.	Status timeline (Not applied / Applied / Screening / Inducted / Suspended); eligibility checklist; fee and validity; renew and upgrade to full member.	Membership, Application
Events	Discover events relevant to the student.	Filter by type, directorate, date and eligibility; seat availability; waitlist; add-to-calendar and reminders.	Event
Event Registration	Convert interest into a confirmed seat.	Eligibility rule check (year, quota, prerequisites); duplicate control; QR e-pass on confirmation; cancellation before cut-off frees the seat.	Registration, Event
Volunteer Opportunities	Open calls for student volunteers.	Opportunity cards (role, shift, skill need, seats); apply with skill match score; accept / decline assignment; service-hours ledger.	VolunteerOpportunity, Application

Module	Purpose	Key functions / rules	Primary data entities
Attendance	Personal attendance record across all activities.	Session-wise record with date, event, method and status; percentage by category; deficiency warning below threshold; self-mark with geo/photo proof where allowed.	Attendance, Session
Certificates	Earned certificates and verification.	Auto-issued on rule fulfilment; PDF download with serial number and QR; verification link; transcript-style consolidated statement.	Certificate, Attendance
Achievements	Recognition beyond attendance.	Badges and milestones (best volunteer, event lead, competition); leaderboard opt-out; publish-to-public-portal consent toggle.	Achievement
Notifications	Timely, relevant communication.	In-app bell, e-mail and SMS channels; preferences per category; digest mode; read receipts.	Notification
Feedback	Capture experience after every activity.	Post-event rating (1-5) with comments; complaint escalation; anonymity option; closure notification when resolved.	Ticket, Feedback

PORTAL 02

Member Portal

Working space of an inducted YPDC member - duties, tasks, performance and reporting

Portal Specification — Member Portal

Purpose

A member is a student who has completed induction and holds a Member ID with a directorate allocation. The Member Portal is transactional: it shows what the member owns (responsibilities, tasks, duties), records what the member did (attendance, activity logs) and reflects how the member is performing (score, report, certificate).

Attribute	Detail
Target users	Inducted members, team leads within a directorate.
Access level	Authenticated. Full read/write on own records; read-only on directorate objectives and work plan.
Modules	10 (inventoried below)

Main user journey

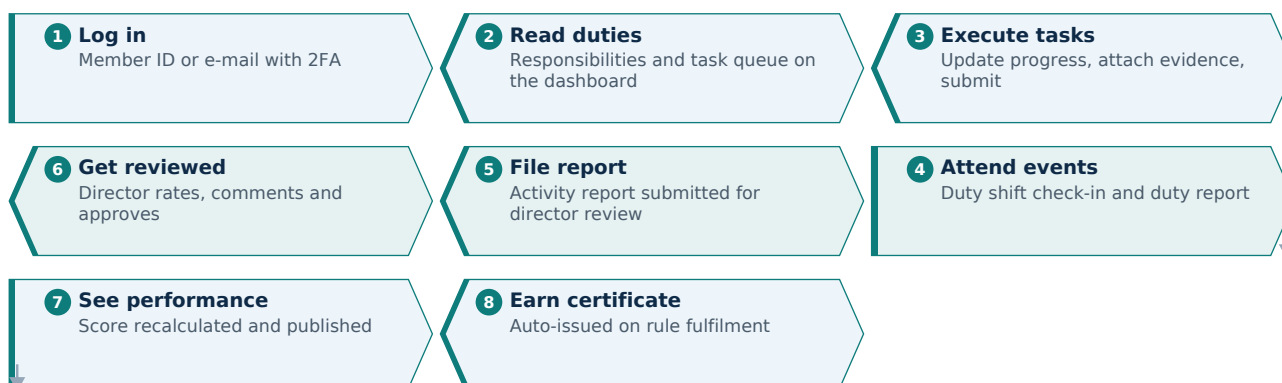


Figure 15 — Member Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
Profile & Member ID	Verified identity and digital membership card.	Auto-generated Member ID (format YPDC-YYYY-DIR-NNNN); QR card for check-in; profile verification badge; editable contact with admin re-verification for critical fields.	User, Profile, Membership
Directorate	The member's organisational home.	Assigned directorate with mandate and leadership; team roster; switching requires director approval; transfer history.	Directorate, DirectorateMember
Responsibilities	Standing duties attached to the member's role.	Role-based responsibility list; acceptance acknowledgement; review at induction and on role change; non-acceptance flagged to director.	Responsibility, DirectorateMember
Tasks	Discrete assignments with owners and deadlines.	Assigned / in-progress / submitted / approved states; file and link attachments; progress updates; overdue alerts to member and director.	Task
Events / Duties	Duty roster for events and activities.	Duty allocation with shift, venue and reporting officer; accept / request swap; duty report submission after the event.	Event, DutyAssignment

Module	Purpose	Key functions / rules	Primary data entities
Attendance	Attendance across sessions, events and duties.	Category-wise percentage; minimum thresholds per category; deficiency notice and condonation request; export for personal record.	Attendance, Session
Certificates	Participation, completion and merit certificates.	Rule-based auto-issue; director recommendation for merit certificates; QR-verified PDF; renewal on repeated participation.	Certificate
Performance	Objective, transparent performance standing.	Score components (tasks 40%, attendance 25%, events 20%, conduct 15%); monthly and cumulative view; director comments; appeal within 7 days.	Performance
Reports	Member-generated activity reporting.	Activity / duty / incident report templates; draft and submit; director acknowledgement; history with reviewer remarks.	ActivityReport
Notifications	Task, duty and approval alerts.	Deadline reminders (T-3, T-1, overdue); duty roster published; approval outcomes; channel preferences.	Notification

PORTAL 03

Directorate Portal

Management console for directors, assistant directors and team leads

Portal Specification — Directorate Portal

Purpose

Directorates are the delivery engine of YPDC. This portal lets a directorate plan (objectives, work plan), organise (team, tasks, duties), execute (events, activities) and prove its work (activity reports, achievements, performance). Every artefact created here feeds the Admin Portal's approvals, attendance and analytics modules.

Attribute	Detail
Target users	Director, Assistant Director, team leads of a directorate.
Access level	Authenticated, scoped to own directorate. Proposes and recommends; final approvals rest with Admin/Faculty per the approval matrix.
Modules	13 (inventoried below)

Main user journey



Figure 16 — Directorate Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
Directorate Profile	Identity and mandate of the unit.	Name, mandate, focus areas, established date, contact; public summary mirrored on the Guest Portal; edit requests routed to Admin.	Directorate
Director / Assistant Director	Leadership record and delegation.	Appointment with tenure; delegation matrix; acting arrangements; handover checklist on change of office.	DirectorateMember, Role
Team Members	Roster management.	Add / remove members with reason; skill and availability matrix; sub-team grouping; capacity limits per activity.	DirectorateMember, User
Objectives	What the directorate must achieve.	SMART objectives linked to the YPDC annual plan; owner, target value and due date; quarterly review status.	Objective
Work Plan	Time-phased breakdown of objectives.	Monthly / quarterly plan lines; milestones; dependency flags; variance against plan; faculty and admin visibility.	WorkPlan, Objective

Module	Purpose	Key functions / rules	Primary data entities
Tasks	Delegation and follow-up.	Create and assign tasks with deadline and priority; sub-tasks; bulk assign to a sub-team; progress dashboard and escalation on delay.	Task
Events & Activities	Delivery calendar of the directorate.	Activity register; owner, venue, budget, expected footfall; checklists (venue, media, logistics); post-activity closure.	Event, Activity
Event Proposals	Formal request to hold an event.	Proposal form (title, rationale, objectives, audience, budget, date, risk); document attachments; multi-level approval chain; status tracking with SLA timer.	EventProposal, ApprovalWorkflow
Attendance	Attendance administration for the unit.	Create session, generate QR / code; mark attendance from device; import manual list; excuse and condonation requests; reconciliation report.	Attendance, Session
Activity Reports	Evidence of work done.	Post-event report (summary, footfall, outcomes, budget used, media links, lessons); director sign-off; faculty review; publication to public portal.	ActivityReport, MediaAsset
Achievements	Unit-level recognition.	Log directorate and member achievements; recommendation to Admin for certificate; public display with consent.	Achievement
Documents	Unit document repository.	SOPs, templates, minutes, reports; folder structure with version control; access level (unit / member / admin); retention and disposal rules.	Document
Performance	Unit and member performance oversight.	Directorate scorecard against objectives; member ranking with justification; low-performer action plan; period comparison.	Performance, Objective

PORTAL 04

Faculty Portal

Academic oversight, endorsement and approval layer between students and administration

Portal Specification — Faculty Portal

Purpose

Faculty members act as focal persons and endorsers. They do not run day-to-day operations; they verify academic legitimacy, approve proposals and recommendations, monitor student and member activity, and provide structured feedback to both the directorate and the administration.

Attribute	Detail
Target users	Faculty focal persons, department heads, faculty advisors.
Access level	Authenticated. Read across assigned directorates; approval and recommendation rights; no user administration.
Modules	9 (inventoried below)

Main user journey

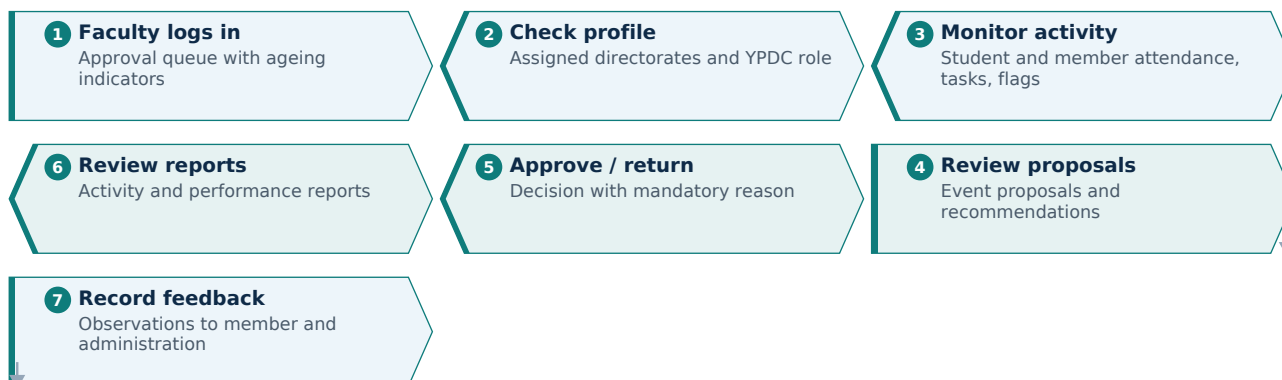


Figure 17 — Faculty Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
Faculty Profile	Academic identity and contact.	Qualification, department, office hours, contact; assigned directorates and student groups; availability for approvals.	User, Profile, FacultyRole
Department / Designation	Institutional position.	Department, designation, tenure; mapping to YPDC scope of oversight; change notified to Admin for permission recalculation.	FacultyRole
YPDC Role	Formal role inside YPDC.	Focal person, advisor, patron or reviewer; scope (which directorates / events); deputy assignment; term dates.	FacultyRole, Role
Events	Events requiring faculty oversight.	Events of assigned directorates with proposal status; faculty comments; attendance as chief guest / judge; evaluation form after the event.	Event, EventProposal
Student / Member Activities	Supervision view of activity records.	Activity feed per student / member; attendance and task summary; risk flags (low attendance, overdue tasks); intervention note.	Attendance, Task, ActivityReport
Approvals	Faculty-level approval queue.	Queue with ageing; approve / return with reason; recommendation for membership, certificates and awards; bulk actions with per-item remarks.	ApprovalWorkflow, Application

Module	Purpose	Key functions / rules	Primary data entities
Reports	Reports requiring academic review.	Activity and performance reports of assigned units; review comments; endorsement or rejection back to the directorate.	ActivityReport, Performance
Feedback	Structured feedback to members and administration.	Member feedback forms; directorate-level observations; policy suggestions; visibility control (to admin only / shared with directorate).	Feedback
Notifications	Approval and review alerts.	New approval SLA alerts; overdue reminders; report submitted; escalation notices.	Notification

PORTAL 05

Admin Portal

Central control room - users, content, approvals, attendance, certificates, analytics and permissions

Portal Specification — Admin Portal

Purpose

The Admin Portal is the only place where the platform itself is configured and where cross-portal decisions are finalised. It owns identity, roles and permissions; it is the final authority in every approval chain; it publishes the public website; and it produces the analytics on which the executive committee reports.

Attribute	Detail
Target users	Super admin, admin officers, content managers, data / analytics officers.
Access level	Authenticated with 2FA. Full rights scoped by role and permission set; every action audit-logged.
Modules	15 (inventoried below)

Main user journey



Figure 18 — Admin Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
Dashboard	Single operational picture.	KPI tiles (members, active events, pending approvals, attendance %, complaints); trend charts; directorate comparison; alerts and to-do queue.	All (read)
User Management	Identity lifecycle for every portal user.	Create / invite / suspend / delete accounts; role assignment; password reset and 2FA enforcement; bulk import from CSV; last-login and status reports.	User, Role
Member Management	Membership lifecycle administration.	Screen and approve applications; issue Member IDs; transfers between directorates; suspension / restoration with reason; renewal and expiry runs.	Membership, Application, Profile
Directorate Management	Structure of the organisation.	Create / merge / retire directorates; appoint directors and assistants; define mandate and capacity; reassign teams.	Directorate, DirectorateMember
Faculty Management	Faculty onboarding and scope.	Register faculty; assign department, designation and YPDC role; map to directorates; revoke on term end.	User, FacultyRole

Module	Purpose	Key functions / rules	Primary data entities
Events Management	Master event register and approvals.	Approve / reject proposals; publish to public calendar; manage capacity, waitlist and cancellation; conflict detection on venue and date.	Event, EventProposal
Applications & Approvals	Unified approval engine.	Configurable multi-level chains per form type; SLA timers with auto-escalation; approve / return / delegate; complete decision history.	Application, ApprovalWorkflow
Attendance	Attendance governance.	Approve sessions created by directorates; audit anomalies (duplicate, out-of-geo, late mark); grant condonation; lock periods for reporting.	Attendance, Session
Certificates	Certificate authority of the platform.	Design templates; approve auto-drafts; issue with unique serial and QR; revoke with reason; public verification endpoint.	Certificate
Reports & Analytics	Insight for decision-making.	Membership funnel; event ROI (cost vs footfall); attendance heat map; directorate scorecards; scheduled exports (PDF / Excel) to stakeholders.	All (aggregate)
Notifications	Broadcast and system messaging.	Targeted broadcast by role, directorate or cohort; templates; scheduling; delivery and read analytics.	Notification
Website Content	Content management for the public portal.	Pages, news, FAQs, projects, gallery albums; rich editor with media library; draft / review / publish with reviewer; SEO fields and versioning.	ContentPage, MediaAsset, Album
Documents	Central document repository.	Policies, SOPs, forms, minutes; classification (public / member / admin); version history; access grants and download audit.	Document
Feedback / Complaints	Grievance handling.	Triage by category and priority; assignment with SLA; resolution notes; complainant satisfaction; monthly trend and root-cause report.	Ticket, Feedback
Roles & Permissions	Access control configuration.	Define roles and permission sets (module x action); assign to users; segregation-of-duty rules; permission change audit and periodic access review.	Role, Permission, UserRole

PORTAL 06

Alumni Portal

Life after induction - network, mentorship, careers and giving back

Portal Specification — Alumni Portal

Purpose

Alumni remain the most valuable asset of YPDC. This portal keeps their profile current, connects them to each other and to current members through mentorship, routes career and volunteer opportunities both ways, and records their continued contribution.

Attribute	Detail
Target users	Graduated members, former office bearers, retired faculty advisors.
Access level	Authenticated. Own profile; network and mentorship by mutual consent; read access to public and alumni events.
Modules	10 (inventoried below)

Main user journey

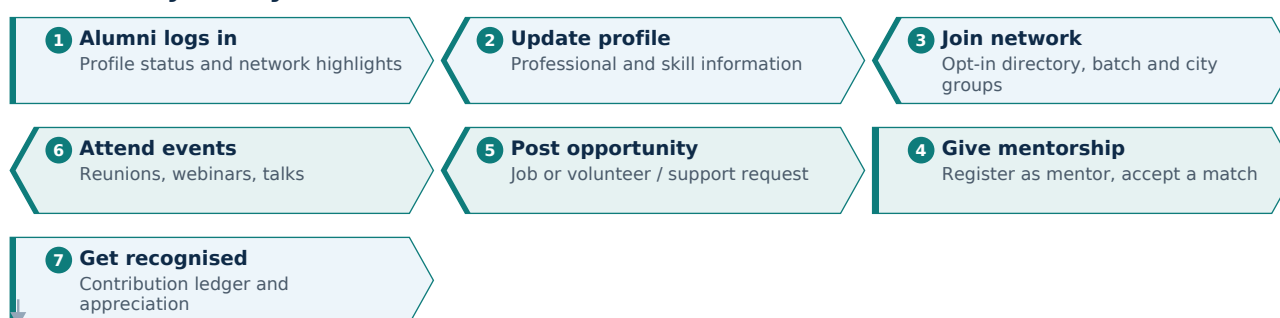


Figure 19 — Alumni Portal: principal user journey through the portal.

Module inventory

Each module below is a screen group in the portal's navigation. “Primary data entities” refers to the data model in Section 9.

Module	Purpose	Key functions / rules	Primary data entities
Alumni Profile	Current identity after graduation.	Batch, Member ID retention, contact, city and country; privacy controls per field; verified alumni badge.	User, Profile, AlumniProfile
Academic Information	Academic record preserved.	Institution, programme, graduation year, CGPA (optional), academic achievements; YPDC tenure and directorates served.	AcademicRecord, Membership
Professional Information	Career footprint.	Current organisation, designation, industry, location; employment history; open-to-opportunity flag; LinkedIn sync (manual).	ProfessionalRecord
Skills	Machine-readable expertise.	Skill tags with proficiency level; endorsement by peers; used for mentorship and volunteer matching.	Skill, AlumniProfile
Alumni Network	Directory and community.	Searchable directory (opt-in); batch and city groups; message request with consent; alumni chapter pages.	AlumniProfile, Network
Mentorship	Structured giving back.	Mentor / mentee registration; matching by field, skill and availability; session log; mid-cycle review; closure with dual feedback.	MentorshipPair, Session

Module	Purpose	Key functions / rules	Primary data entities
Events	Alumni-focused and open events.	Reunions, webinars, talks; alumni-only and open categories; registration and speaking-interest expression.	Event, Registration
Achievements	Professional recognition.	Awards, publications, certifications; YPDC alumni award nominations; consent-based public showcase.	Achievement
Career Opportunities	Two-way job board.	Alumni-posted openings; member applications with referral tag; YPDC internships; posting moderation by admin.	JobPosting, Application
Volunteer / Support Opportunities	Non-financial and financial support.	Guest speaker, judge, trainer, donor and in-kind support requests; commitment tracking; contribution ledger and appreciation certificate.	VolunteerOpportunity, Contribution

6. Data Model Outline

All seven portals read from and write to the entities below. This is a logical model intended to guide detailed design, not a physical schema; field types, indexes and constraints are decided in the design phase.

6.1 Core entities

#	Entity	Key attributes	Relationships	Written by
1	User	id, full name, e-mail, phone, password hash, 2FA flag, status, created / last login	1-n UserRole, 1-1 Profile	Admin / system
2	Profile	photo, date of birth, address, institution, programme, verification status	1-1 User	User (self), Admin verifies
3	Role	code, name, description, portal, permission set	1-n UserRole, 1-n Permission	Admin
4	Permission	module, action (create / read / update / delete / approve)	n-n Role	Admin
5	UserRole	user, role, scope (directorate / department), valid from / to	n-1 User, n-1 Role, n-1 Directorate	Admin
6	Membership	member ID, tier, status, join date, validity, directorate, dues status	1-1 User, n-1 Directorate	Admin
7	Application	type (membership / volunteer / other), payload, documents, reference ID, status, decision	n-1 User, 1-1 ApprovalWorkflow	Guest / Student, Admin
8	Directorate	name, mandate, focus areas, established date, status, public summary	1-n DirectorateMember, 1-n Objective	Admin
9	DirectorateMember	directorate, member, role (director / assistant / lead / member), from / to, reason	n-1 Directorate, n-1 User	Director / Admin
10	Objective	title, description, owner, target value, due date, linked plan, status	n-1 Directorate, 1-n WorkPlan	Director
11	WorkPlan	period, activity line, owner, milestone dates, status, variance	n-1 Objective	Director
12	Task	title, description, assignee, assigner, due date, priority, status, evidence, rating	n-1 Directorate, n-1 User	Director / Member
13	Event	title, type, date, venue, capacity, status, budget, owner directorate, publish flag	n-1 Directorate, 1-1 EventProposal, 1-n Registration, 1-n Session	Directorate / Admin
14	EventProposal	rationale, objectives, audience, budget, date, risk, documents, status	1-1 Event, 1-1 ApprovalWorkflow	Directorate
15	Registration	event, user, status (confirmed / waitlist / cancelled), e-pass code, checked in	n-1 Event, n-1 User	User / system
16	DutyAssignment	event, member, role, shift, venue, reporting officer, acceptance, duty report	n-1 Event, n-1 User	Directorate
17	Session	title, event, date, start / end, venue, geo, marking method, marking window, lock status	n-1 Event, 1-n Attendance	Directorate
18	Attendance	session, user, status, timestamp, method, geo, device, marked by, condonation	n-1 Session, n-1 User	Directorate / system
19	VolunteerOpportunity	title, role, skills needed, shifts, seats, location, window, status	n-1 Directorate, 1-n Application	Directorate / Admin
20	Certificate	type, user, event, template, serial, QR token, issued by, issue date, status	n-1 User, n-1 Event	Admin / system
21	Achievement	title, type, user or directorate, date, awarding body, publish flag, consent	n-1 User or Directorate	Directorate / Admin
22	Performance	user, period, component scores, total, director comment, adjustment, appeal status	n-1 User, n-1 Directorate	System / Director
23	ActivityReport	event or directorate, period, summary, outcomes, footfall, budget used, media, status	n-1 Event or Directorate	Member / Directorate
24	Ticket / Feedback	type, category, priority, subject, body, anonymity flag, owner, SLA, status, resolution, rating	n-1 User (optional), 1-n TicketNote	Any user / Admin

#	Entity	Key attributes	Relationships	Written by
25	Notification	template, recipient, channel, payload, status, sent / read timestamps	n-1 User	System / Admin
26	ContentPage	slug, title, body, type (page / news / FAQ / project), SEO fields, status, version, owner	1-n MediaAsset	Content Manager
27	MediaAsset / Album	file, type, size, album, caption, consent flag, credit, publish flag	n-1 Album, n-1 ContentPage or Event	Content Manager
28	Document	title, category, file, version, access level (public / member / admin), retention, uploaded by	n-1 Directorate (optional)	Admin / Directorate
29	ApprovalWorkflow	form type, chain definition, current level, status, SLA due, delegation	1-1 source record, 1-n ApprovalStep	System
30	ApprovalStep	level, approver, action, remark, decided at, time taken	n-1 ApprovalWorkflow	Approver / system
31	AlumniProfile	batch, graduation year, city, country, organisation, designation, open-to-opportunity	1-1 User, 1-n Skill	Alumni (self)
32	MentorshipPair	mentor, mentee, field, term, cadence, goals, status, closure feedback	n-1 User (x2), 1-n Session	Alumni lead / system
33	JobPosting	title, organisation, type, description, posted by, status, applications	n-1 User (alumni), 1-n Application	Alumni / Admin
34	Contribution	type (mentorship / speaker / donor / in-kind), user, hours or value, event, recognised	n-1 User	System / Admin
35	AuditLog	actor, action, entity, entity id, before / after, IP, user agent, timestamp	immutable, append-only	System

6.2 Data rules that apply across all entities

- **Soft delete:** records are never physically removed; a status flag and a deletion reason are stored instead.
- **Immutable audit:** every create, update, approve and delete on a regulated entity writes an AuditLog entry that cannot be edited.
- **Consent flags:** any personally identifiable record that may be shown publicly carries an explicit, revocable consent flag.
- **Retention:** academic and membership records are retained for the statutory period; media and logs follow the retention schedule in Section 10.
- **Ownership:** every record stores who created it and who last modified it, and these fields cannot be overwritten by a user.

7. Security, Privacy & Access Control

7.1 Authentication & session control

Control	Specification
Account creation	Admin-created or self-service with e-mail verification; institutional e-mail preferred where available.
Password policy	Minimum 10 characters, complexity check, breach-list check, no reuse of the last 5 passwords.
Second factor	OTP by e-mail or SMS for all staff roles (Director and above, Faculty, Admin); optional for members.
Session	Idle timeout 30 minutes (staff) / 8 hours (members); concurrent-session limit; "sign out all devices" available to the user.
Lockout	5 failed attempts → 15-minute lockout with notification; pattern-based attempts blocked.
Password reset	Self-service via time-limited single-use token; never sent in plain text; admin reset requires a recorded reason.
Device audit	New-device login recorded with IP, user agent and location, and notified to the user.

7.2 Authorisation & segregation of duty

- Permissions are evaluated **server-side on every request**; hiding a button in the interface is not an access control.
- Row-level scope is enforced for directorate and faculty roles — a director cannot read another directorate's records even by editing a URL.
- A user may not approve a workflow they initiated, and may not modify their own role or permission set.
- Permission changes are effective immediately, audit-logged, and reviewed in a periodic access review each term.
- Bulk exports are restricted to Admin and Super Admin and are watermarked with the requesting user's identity.

7.3 Data protection & privacy

Area	Position
Legal basis	Consent is captured at every public form, with a plain-language statement of purpose and retention.
Data minimisation	Only fields required for a stated purpose are collected; optional fields are clearly marked.
Anonymity	Complaints may be filed anonymously; the system stores no identifier once anonymity is selected.
Public exposure	Member names, photos and achievements appear publicly only where a consent flag is set.
Third parties	No personal data is shared with third parties except where required by law, with a recorded decision.
Subject rights	A member may request access to, correction of, or export of their own data from their profile page.
Retention	Membership and academic records: statutory period. Media: 5 years. Audit logs: 3 years. Session logs: 90 days.
Breach response	Defined incident procedure with notification obligations and a named responsible officer.

7.4 Technical controls

- HTTPS enforced everywhere; HTTP redirected; HSTS enabled; TLS 1.2 or higher.
- Protection against injection, cross-site scripting, CSRF and insecure direct object references (IDOR) as standard practice, verified by test.
- Uploaded files scanned, type-restricted and served from a separate, non-executable storage location.
- Rate limiting on authentication and public forms to blunt credential stuffing and spam.
- Daily encrypted backups with a tested restore procedure; separate staging and production environments.

- Centralised logging and alerting; monitoring of error rates, response times and failed-login spikes.

8. Non-Functional Requirements

Category	Requirement	Target	How it is verified
Performance	Page load for authenticated portal pages on a 4G connection	Under 2.5 seconds for first contentful paint	Lighthouse / field measurement at acceptance
Performance	API response for standard list and detail endpoints	p95 under 400 ms	Load test report
Capacity	Concurrent users during a large event check-in window	1,000 concurrent with no degradation	Load test at 2× target
Availability	Platform availability during working hours	99.5% monthly	Monitoring dashboard, monthly report
Scalability	Growth headroom without architectural change	10× current record volume	Design review + capacity test
Accessibility	Usable by people with common impairments	WCAG 2.1 AA on public pages	Audit before public launch
Usability	A new member can register for an event unaided	Under 60 seconds from login	Usability test with 5 users
Compatibility	Browsers and devices	Latest two versions of Chrome, Edge, Safari, Firefox; Android and iOS browsers	Cross-browser test matrix
Language	Interface language support	English first; Urdu strings prepared for later localisation	String externalisation review
Maintainability	Adding a new approval form type	By configuration only, no code release	Demonstration at acceptance
Observability	Detecting a production incident	Alert within 5 minutes of an error-rate spike	Fault injection test
Portability	Data export	Full export of any entity in CSV/Excel by an authorised admin	Demonstration at acceptance

9. Delivery Roadmap

The platform is delivered in six phases. Each phase ends with something that can actually be used, so that value and feedback arrive early rather than at the end. Durations assume a small dedicated team and are indicative pending confirmation of resources.

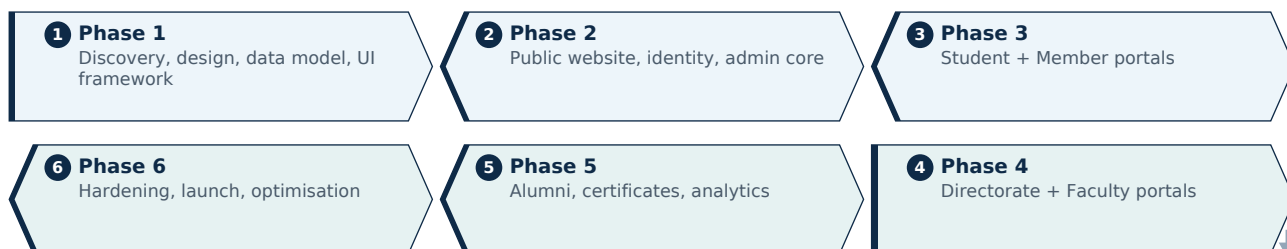


Figure 20 — Six-phase delivery sequence. Phases 2 and 3 already deliver a usable product.

Phase	Duration	Scope	Key deliverables	Acceptance criteria
1. Foundation	4–6 weeks	Requirements confirmation, information architecture, data model, design system, environment setup	Signed-off SRS, ERD, clickable prototype, CI/CD pipeline, staging environment	Executive Committee approves scope; prototype walked through by directors and faculty
2. Public + Admin core	6–8 weeks	Guest Portal (14 modules), authentication, user and role management, CMS, notifications	Live public website, admin user administration, content publishing	Content team publishes a page, news item and gallery album without developer help
3. Student + Member	6–8 weeks	Student (10) and Member (10) portals, membership workflow, attendance service, basic certificates	Working onboarding, event registration, attendance marking, certificate issue	One real event run end-to-end on the platform with attendance and certificates
4. Directorate + Faculty	6–8 weeks	Directorate (13) and Faculty (9) portals, approval engine, performance, activity reports	Full approval chains, work plan and tasks, performance scoring	An event proposal approved through all four levels inside the SLA
5. Alumni + analytics	4–6 weeks	Alumni (10) portal, mentorship, career board, reporting and analytics dashboards	Alumni network, mentorship matching, executive dashboard and scheduled exports	Quarterly executive report generated entirely from system data
6. Hardening & launch	4 weeks	Load and security testing, accessibility audit, data migration, training, go-live	Test reports, training material and sessions, migration report, production launch	Load test at 2× target passed; security findings closed; 80% of directors trained

Sequencing logic

Public website first because it is visible, low-risk and immediately useful for applications.

Attendance before performance because performance scoring depends on trustworthy attendance data.

Approvals before directorate planning so that the directorate portal opens into a working approval chain rather than a dead end.

Alumni last because it depends on member data that must already be clean.

10. Success Measures & KPIs

Area	KPI	Definition	Target (12 months after launch)	Reported to
Adoption	Active member rate	Members logging in at least once in 30 days ÷ total members	70%	Executive Committee, monthly
Adoption	Portal-driven applications	Applications submitted online ÷ total applications	95%	Admin, monthly
Efficiency	Approval cycle time	Median working days from submission to final decision	Under 5 days	Executive Committee, monthly
Efficiency	SLA compliance	Approvals decided within SLA ÷ total approvals	90%	Admin, weekly
Operations	Event closure rate	Events with a submitted activity report within SLA ÷ events held	95%	Directors, monthly
Operations	Attendance coverage	Attendee records captured ÷ confirmed registrations	90%	Directors, per event
Quality	Attendance disputes	Disputes raised per 1,000 attendance marks	Under 10	Admin, monthly
Quality	Complaint resolution	Tickets resolved within SLA ÷ tickets received	85%	Executive Committee, monthly
Quality	Complainant satisfaction	Average satisfaction rating on closed tickets (1-5)	4.0+	Admin, monthly
Recognition	Certificate turnaround	Median days from event closure to certificate issue	Under 2 days	Admin, monthly
Recognition	Certificate verifications	Public verification lookups per month	Growing trend	Admin, quarterly
Engagement	Volunteer conversion	Volunteers deployed ÷ volunteer applications received	60%	Directorates, per event
Engagement	Alumni mentorship pairs	Active mentorship pairs at period end	25+	Alumni lead, quarterly
Engagement	Public site reach	Unique public visitors per month	Growing trend	Content Manager, monthly
Platform health	Uptime	Availability during working hours	99.5%	Technical lead, monthly
Platform health	Incidents	Severity-1 incidents per quarter	0	Technical lead, quarterly

11. Risks & Mitigations

#	Risk	Likelihood	Impact	Mitigation	Owner
R1	Personal data of members or students is exposed publicly	Medium	High	Consent flags on every publishable record; default to private; access review each term; publication requires a second reviewer.	Admin / Technical lead
R2	Low adoption — members and directors keep using spreadsheets	High	High	Phase-wise rollout with champions in each directorate; make the platform the only route to certificates and approvals; short training sessions.	Project Sponsor
R3	Attendance marking is abused or disputed	Medium	High	Time-boxed QR, geo checks, marker identity, reconciliation before lock, admin sampling and a documented condonation route.	Director / Admin
R4	Approval bottlenecks at the Admin level	High	Medium	SLA timers with auto-escalation, delegation on leave, threshold-based routing, and a weekly pending-approvals report.	Admin
R5	Performance scoring is perceived as unfair	Medium	High	Published weights, visible components, mandatory justification for manual adjustment, 7-day appeal window, exclusion of unassigned periods.	Director / Faculty
R6	Scope creep delays the launch	High	Medium	This document as the agreed baseline; a formal change-request route; new requests logged for a later phase by default.	Project Sponsor
R7	Content on the public site becomes stale or inaccurate	Medium	Medium	Named owner per page, review-by date, quarterly content audit, last-updated stamp visible publicly.	Content Manager
R8	Single-person dependency on the technical team	Medium	High	Documentation, standard framework, code repository with review, and a support agreement covering the first year.	Technical lead
R9	Hosting, cost or connectivity constraints	Medium	Medium	Hosting decision in Phase 1; responsive design tested on low bandwidth; offline code fallback for attendance.	Admin
R10	Data migration from existing registers is incomplete	Medium	Medium	Migration plan with a data-cleaning sprint, duplicate resolution rules, and a parallel-run period before old registers are retired.	Admin

12. Assumptions & Open Items

12.1 Assumptions on which this specification rests

- The abbreviation **YPDC** and the seven-portal structure reflect the intended organisational model; the full legal name is to be confirmed.
- Directorates, directors, assistant directors and faculty focal persons exist as described, and appointments are recorded formally.
- Members have access to a smartphone or shared device capable of running a modern browser.
- An administrative team is available to operate the Admin Portal on a routine basis.
- Membership fees, where applicable, are collected outside this system in this phase.
- Existing paper and spreadsheet records are available for migration in a readable form.
- Hosting, domain, e-mail and SMS gateway arrangements will be confirmed before Phase 2 begins.

12.2 Open items requiring a decision

#	Open item	Options	Recommendation	Needed by
D1	Full legal name of YPDC	Confirm the expansion of the abbreviation	Confirm and substitute globally before external circulation	Before Phase 2
D2	Attendance marking method	QR / geo-fence / offline code / manual	Enable all four; make QR the default	Phase 1
D3	Membership fee model	Free / annual fee / tiered	Decide before the membership workflow is configured	Phase 2
D4	Performance weighting	As proposed (40/25/20/15) or revised	Adopt as proposed for the first two periods, then review with data	Phase 4
D5	Hosting	Institutional server / managed cloud / hybrid	Managed cloud with daily backups, subject to policy approval	Phase 1
D6	Urdu localisation	Now or after stabilisation	Externalise strings now; translate after Phase 3	Phase 2
D7	Certificate signatories	Who signs each certificate type	Confirm the signatory matrix before certificate templates are built	Phase 3
D8	Grievance committee composition	Members, chair, quorum, P1 handling	Formally constitute before the complaint workflow goes live	Phase 3

13. Appendix A — Complete Module Inventory (Sitemap)

The authoritative list of all 81 modules across the seven portals, in the order they will appear in each portal's navigation.

Ref	Module	Portal	Purpose
07.1	About YPDC	Guest / Public Portal	Introduce the organisation, its history, mandate and structure.
07.2	Vision & Mission		Publish strategic intent and core values.
07.3	Leadership		Show governance and patronage.
07.4	Directorates		Present all directorates as functional units.
07.5	Projects		Showcase ongoing and completed projects.
07.6	Events		Public event calendar and event detail pages.
07.7	News & Updates		Announcements, press releases and notices.
07.8	Achievements		Celebrate organisational and member achievements.
07.9	Gallery		Visual record of activities.
07.10	Publications		Repository of reports, newsletters and research.
07.11	Membership		Explain membership and accept applications.
07.12	Volunteer Registration		Capture public volunteers for YPDC activities.
07.13	Contact Us		Single entry point for enquiries.
07.14	FAQs		Self-service answers to repeated questions.
01.1	Profile	Student Portal	Single source of truth for student identity and contact.
01.2	Membership		Show and progress the student's membership status.
01.3	Events		Discover events relevant to the student.
01.4	Event Registration		Convert interest into a confirmed seat.
01.5	Volunteer Opportunities		Open calls for student volunteers.
01.6	Attendance		Personal attendance record across all activities.
01.7	Certificates		Earned certificates and verification.
01.8	Achievements		Recognition beyond attendance.
01.9	Notifications		Timely, relevant communication.
01.10	Feedback		Capture experience after every activity.
02.1	Profile & Member ID	Member Portal	Verified identity and digital membership card.
02.2	Directorate		The member's organisational home.
02.3	Responsibilities		Standing duties attached to the member's role.
02.4	Tasks		Discrete assignments with owners and deadlines.
02.5	Events / Duties		Duty roster for events and activities.
02.6	Attendance		Attendance across sessions, events and duties.
02.7	Certificates		Participation, completion and merit certificates.
02.8	Performance		Objective, transparent performance standing.
02.9	Reports		Member-generated activity reporting.
02.10	Notifications		Task, duty and approval alerts.
03.1	Directorate Profile	Directorate Portal	Identity and mandate of the unit.
03.2	Director / Assistant Director		Leadership record and delegation.

Ref	Module	Portal	Purpose
03.3	Team Members	Faculty Portal	Roster management.
03.4	Objectives		What the directorate must achieve.
03.5	Work Plan		Time-phased breakdown of objectives.
03.6	Tasks		Delegation and follow-up.
03.7	Events & Activities		Delivery calendar of the directorate.
03.8	Event Proposals		Formal request to hold an event.
03.9	Attendance		Attendance administration for the unit.
03.10	Activity Reports		Evidence of work done.
03.11	Achievements		Unit-level recognition.
03.12	Documents		Unit document repository.
03.13	Performance		Unit and member performance oversight.
04.1	Faculty Profile	Admin Portal	Academic identity and contact.
04.2	Department / Designation		Institutional position.
04.3	YPDC Role		Formal role inside YPDC.
04.4	Events		Events requiring faculty oversight.
04.5	Student / Member Activities		Supervision view of activity records.
04.6	Approvals		Faculty-level approval queue.
04.7	Reports		Reports requiring academic review.
04.8	Feedback		Structured feedback to members and administration.
04.9	Notifications		Approval and review alerts.
05.1	Dashboard	Alumni Portal	Single operational picture.
05.2	User Management		Identity lifecycle for every portal user.
05.3	Member Management		Membership lifecycle administration.
05.4	Directorate Management		Structure of the organisation.
05.5	Faculty Management		Faculty onboarding and scope.
05.6	Events Management		Master event register and approvals.
05.7	Applications & Approvals		Unified approval engine.
05.8	Attendance		Attendance governance.
05.9	Certificates		Certificate authority of the platform.
05.10	Reports & Analytics		Insight for decision-making.
05.11	Notifications		Broadcast and system messaging.
05.12	Website Content		Content management for the public portal.
05.13	Documents		Central document repository.
05.14	Feedback / Complaints		Grievance handling.
05.15	Roles & Permissions		Access control configuration.
06.1	Alumni Profile		Current identity after graduation.
06.2	Academic Information		Academic record preserved.
06.3	Professional Information		Career footprint.

Ref	Module	Portal	Purpose
06.4	Skills		Machine-readable expertise.
06.5	Alumni Network		Directory and community.
06.6	Mentorship		Structured giving back.
06.7	Events		Alumni-focused and open events.
06.8	Achievements		Professional recognition.
06.9	Career Opportunities		Two-way job board.
06.10	Volunteer / Support Opportunities		Non-financial and financial support.

Total modules

This specification covers **81 modules** across **7 portals**, documented in **11 cross-portal workflows** and **20 figures**.

14. Appendix B — Glossary

Term	Meaning in this document
Assistant Director	Deputy head of a directorate, with delegated authority within the unit.
Approval chain	The configured sequence of approvers a form must pass through, with SLA per level.
Condonation	A formally approved excuse for a shortfall in attendance, recorded with a reason and evidence.
Directorate	A functional unit of YPDC with its own mandate, leadership, objectives and team.
Duty	An assigned role at a specific event or activity, with a shift and a reporting officer.
Faculty focal person	The faculty member responsible for academic oversight of one or more directorates.
Geo-fence	A virtual perimeter around a venue used to validate a self-marked attendance.
Member ID	The permanent unique identifier issued on induction, formatted YPDC-YYYY-DIR-NNNN.
Period lock	The point after which attendance for a period can no longer be changed without an admin adjustment.
Portal	A permission-filtered view of the platform for a specific audience.
Reconciliation	The review step where flagged attendance marks are confirmed or corrected before locking.
Segregation of duty	The rule that a person cannot both initiate and approve the same request.
SLA	Service level agreement — the maximum time allowed for a step before escalation.
Soft delete	Marking a record inactive instead of removing it, preserving history.
Work plan	The time-phased breakdown of a directorate's objectives into activities and milestones.
YPDC	The organisation; the full legal name is to be confirmed by the Executive Committee (see D1).

End of document. This workflow specification is issued as version 1.0 for the approval of the Executive Committee. On approval it becomes the baseline against which all change requests are assessed.